



IISER Mohali

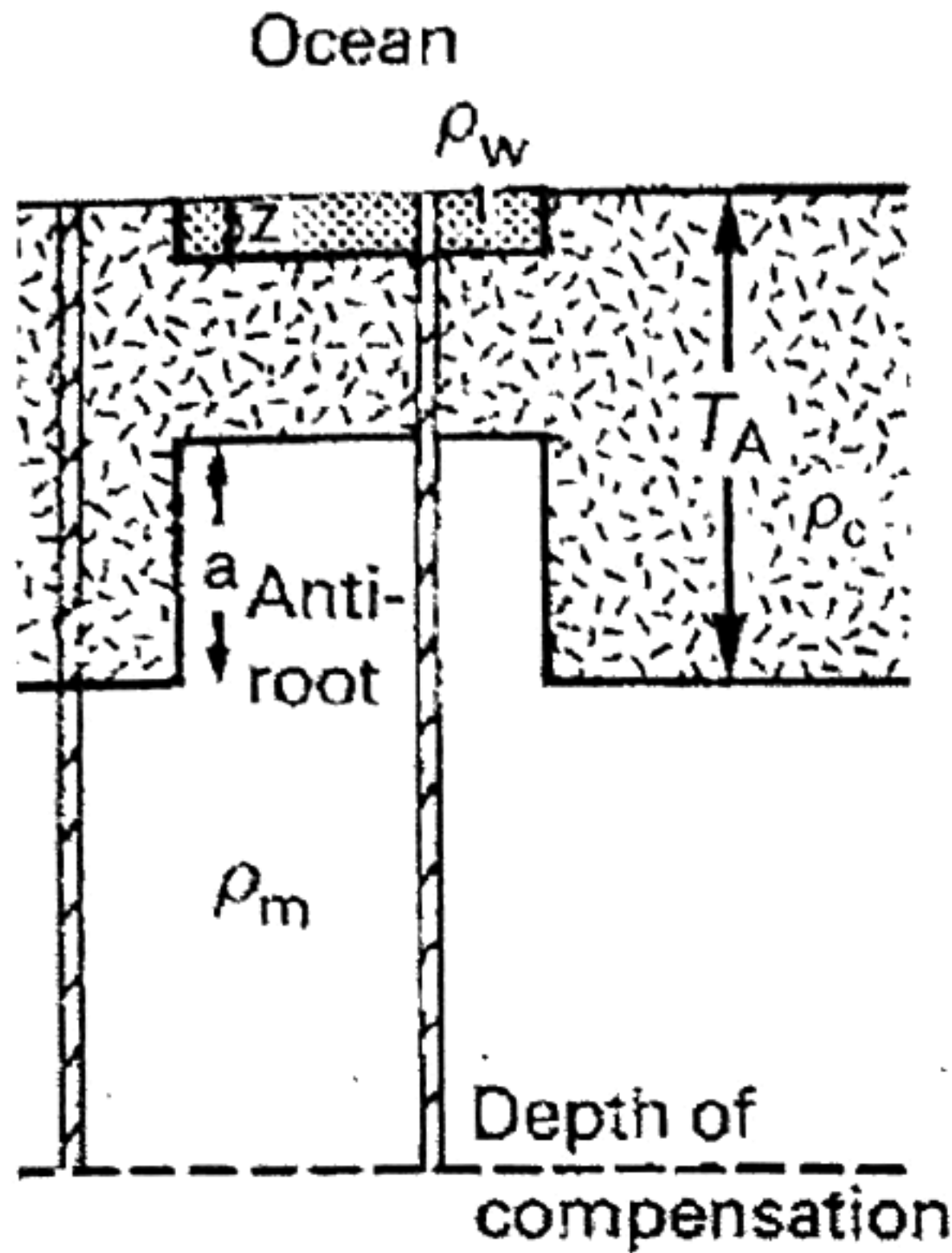
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INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI
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2nd mid semester EES 640 Global Tectonics

Date: 06/03/2026

Instructor: Sharmila Bhattacharya

Total Marks: 25

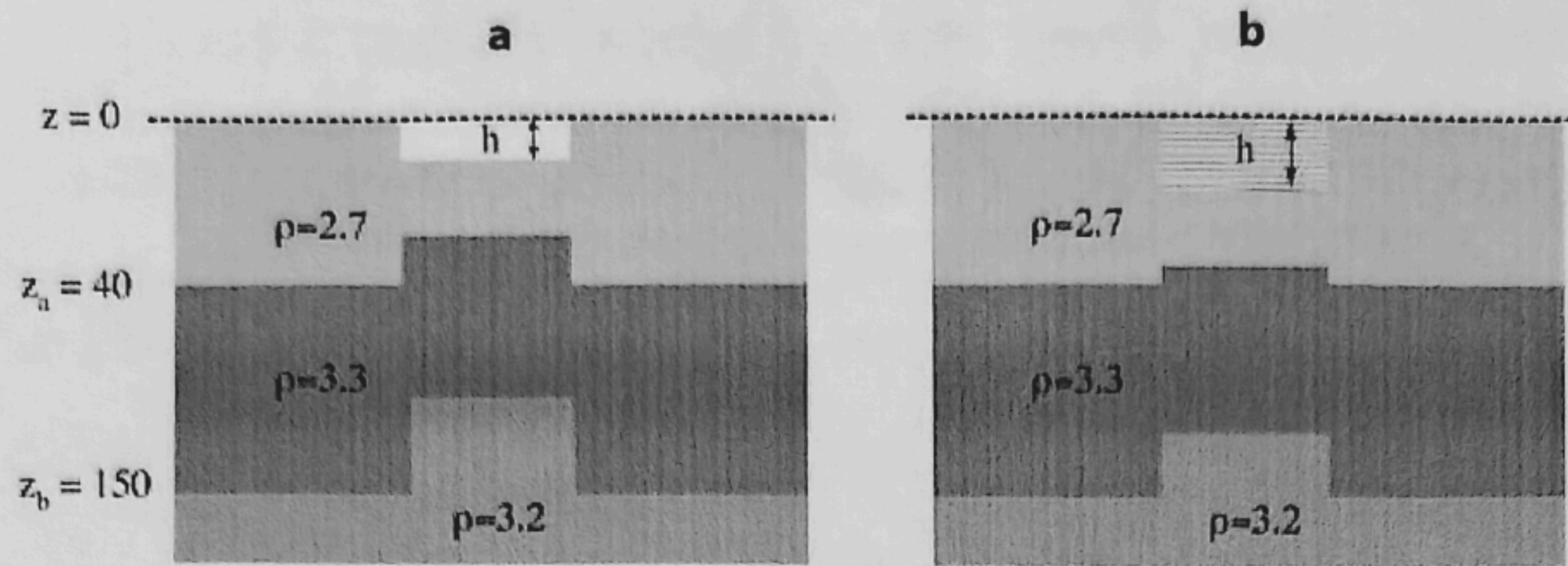


- 1) (a) Refer to the above image to find the condition for isostatic equilibrium in an ocean basin. z , depth of water of density ρ_w ; T_A , normal thickness of crust of density ρ_c ; a , thickness of antiroot; ρ_m , density of mantle. (5)

(b) Consider the problem of continental rifting and rift basin formation. Suppose that 150-km-thick continental lithosphere, consisting of 40-km-thick crust and 110-km-thick lithospheric mantle (image given below), has been extended by 50% so that the crust and lithospheric mantle are both now 0.5 times their original thicknesses. In addition, suppose that the elevation of the surface was initially at sea level. Assume that the density of the air is 0, the density of the crust is 2.7 gm/cm³, the density of the lithospheric mantle is 3.3 gm/cm³, the density of the asthenospheric mantle beneath the lithosphere is 3.2 gm/cm³, and that the rift basin is not filled with either water or sediments.

(i) Determine the depth of the basin relative to sea level. (5)

(ii) Now assume that the basin is filled by sediments with a density of 2.5 gm/cm³. How deep does the basin become with this additional load? (5)



2) True or false: (5)

- (i) Oldest parts of the oceanic crust are found in regions closest to the spreading centers.
- (ii) The brittle-ductile transition occurs at shallow depths in case of ultraslow spreading ridges.
- (iii) Magnetic lineations in the oceans are strong evidence of seafloor spreading.
- (iv) Deep focus earthquakes are associated with constructive plate margins.
- (v) Hotspots are typically associated with intraplate volcanism.

3) Multiple choice questions: (5)

- (i) The ocean ridge basalts are primarily composed of
 - (a) Transitional basalt
 - (b) Calc alkaline basalt
 - (c) Tholeiitic basalt
 - (d) Alkali basalt rich in Na and K
- (ii) Transform plate boundaries are
 - (a) Destructive plate margins
 - (b) Conservative plate margins
 - (c) Constructive plate margins
 - (d) Cannot be defined
- (iii) The core-mantle boundary is at
 - (a) 2891 km
 - (b) 2981 km
 - (c) 5150 km
 - (d) 660 km
- (iv) The past distribution of landmasses forming the Gondwanaland are evidenced by
 - (a) *Glossopteris* flora
 - (b) *Mesosaurus* fauna
 - (c) Permo-Carboniferous glacial deposits
 - (d) All of the above
- (v) Thermoremanent magnetization is acquired
 - (a) Below the Curie temperature
 - (b) Above the Curie temperature
 - (c) There is no relevance of the Curie temperature
 - (d) When magma reaches 100°C